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Car tyres shed a quarter of all microplastics in the environment – urgent action is needed

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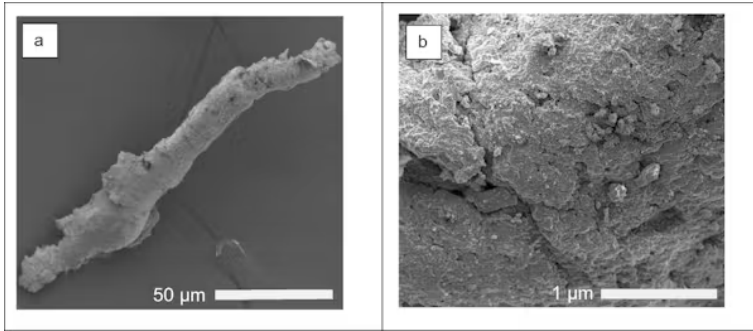
<https://theconversation.com/car-tyres-shed-a-quarter-of-all-microplastics-in-the-environment-urgent-action-is-needed-244132>

Every year, billions of vehicles worldwide shed an estimated [6 million tonnes](#) of tyre fragments. These tiny flakes of plastic, generated by the wear and tear of normal driving, eventually accumulate in the soil, in rivers and lakes, and even in our food. Researchers in [South China](#) recently found tyre-derived chemicals in most human urine samples.

These tyre particles are a significant but often-overlooked contributor to microplastic pollution. They account for [28% of microplastics](#) entering the environment globally.

Despite the scale of the issue, tyre particles have flown under the radar. Often lumped in with other microplastics, they are rarely treated as a distinct pollution category, yet their unique characteristics demand a different approach.

We urgently need to classify tyre particles as a unique pollution category. In our recent [international study](#), colleagues and I found that this approach would drive more focused research that could inform policies specifically designed to mitigate tyre pollution. And it could help ordinary people better understand the scale of the problem and what they can do about it.



Tyre particle pollution under the microscope. (For reference 50 µm (micrometres) is a bit less than the width of a typical human hair). Knight et al (2020) Environmental Science and Pollution Research, CC BY-SA

Right now, delegates are meeting in South Korea to negotiate the first global plastics pollution treaty. While this landmark agreement is poised to address many aspects of plastic pollution, tyre particles are barely on the agenda. Given their significant contribution to microplastics, recognising tyre pollution as a unique issue could help unlock targeted solutions and public awareness. This is what we need to address this growing environmental threat.

Hundreds of chemical additives

Tyre particles tend to be made from a complex mix of synthetic and natural rubbers, along with hundreds of chemical additives. This means the consequences of tyre pollution can be unexpected and far reaching.

For instance, zinc oxide accounts for around 0.7% of a tyre's weight. Though it is essential for making tyres more durable, zinc oxide is highly toxic for fish and other aquatic life and disrupts ecosystems even in trace amounts.



Coho salmon seem to be particularly affected by car tyre chemicals. BLM Oregon / flickr

Another harmful additive is a chemical known as 6PPD, which protects tyres from cracking. When exposed to air and water, it transforms into 6PPD-quinone, a compound linked to mass fish die-offs in the US.

Heavy vehicles, more pollution

We know that heavier vehicles, including electric cars (which have very heavy batteries), wear down their tyres faster and generate more microplastic particles. Car industry experts [Nick Molden and Felix Leach](#) say that, as weight is so crucial to a vehicle's environmental impact, manufacturers should be targeted with weight-based taxes under a "polluter pays" principle. This could encourage lighter vehicle designs while motivating consumers to make greener choices.



As cars get heavier, they tend to shed more tyre particles. Naypong Studio / shutterstock

There are many [questions we still need to investigate](#). For instance, we still don't know how far these tyre particles disperse, or exactly where they are accumulating. To assess their full ecological impact, we need more detailed information on which tyre additives are most toxic, how they behave in the environment, and which species are most at risk ([some salmon species](#) are more sensitive to 6PPD-quinone than others, for example).

In the longer-term, standardised methods will be crucial to measure tyre particles and create effective regulations.

We need global action

Regulatory frameworks, such as the EU's upcoming [Euro 7 emissions standard](#) (which targets vehicle emissions), provide a starting point for controlling tyre emissions. But additional measures are needed.

Innovations in tyre design, such as eco-friendly alternatives to zinc oxide and other materials like 6PPD, could significantly reduce environmental harm. Establishing a [global panel of scientific and policy experts](#), similar to ones that already exist for climate science (known as the IPCC) or biodiversity (IPBES), could further coordinate research and regulatory efforts.

Crucially, we must classify tyre particles as a distinct pollution category. Compared to conventional microplastics, tyre particles behave differently in the environment, break down into unique chemical compounds, and present distinct toxicological challenges.

With more than [2 billion tyres](#) produced each year to fit ever-heavier and more numerous cars, the problem is set to escalate. The environmental toll will only increase unless we recognise and target the specific problem.

Measures like weight-based taxation and eco-friendly tyre innovations would not only reduce tyre pollution but also pave the way for more sustainable transportation systems. The question isn't whether we can afford to act. It's whether we can afford not to.

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